

Supplementary Material: Task-Relevant Ultrasonic Information Acquisition for Impacted Composites

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1 S1. Evidence authority and statistical units

The supplement exposes detailed controls, sensitivities, implementation diagnostics, and machine-readable provenance omitted from the compressed main paper. Numerical authority is the 42-row canonical metrics file. Physical specimens are the first statistical unit and the six held-out experimental domains are equally weighted. State, checkpoint, candidate, seed, and learner- pair rows are repeated computational records rather than independent samples. All teacher, oracle, and component-substitution quantities are retrospective unless an evidence record explicitly marks them deployable.

The codebase refers to the supervised state-conditioned closed-loop implementation as MAVIS. This is one implementation of State-Conditioned Task-Oriented Acquisition, not the paper-level method. The code identifier `mvd_m1_o2` denotes the static deployable reference used in the final comparison; it is not a published competing method.

Record group	Content
S01–S04	Full domain and checkpoint records for spatial, sparse, static, and conditional-value comparisons.
S05–S09	Real-state, acquired-position/history, legacy reconstruction-derived, shuffled-content, and dynamic-valuation controls.
S10–S12	Component substitutions and bounded reachable-set planning records.
S13–S16	Learner-pair value stability, feedback strata, and implementation diagnostics.
S17–S18	Source hashes, chronology, and external-data feasibility boundaries.

2 S2. Spatial information and sparse recoverability

The registered main comparison uses the matched B-family scalar-to-spatial path. The distinct independent internal-field estimator (`U1_INDEPENDENT_FIELD_SENSITIVITY`) had point MAE 0.08964 and is retained only as a sensitivity path. It is not substituted for the matched 32.1% effect.

The sparse condition observes a registered 25% normalized-raster pattern and uses bilinear interpolation before the frozen field pathway. Its 89.9% retention is referenced to the matched full-field gain. The residual sparse-to- full MAE gap remains positive in all domains, and the sampling fraction has no scanner-time interpretation.

3 S3. Task-conditioned and predictor-conditioned value

The main paper uses the preregistered `appearance_oracle` as its task-agnostic C-scan appearance-saliency reference. It assigns a candidate the mean absolute RGB deviation of newly revealed samples from the specimen-specific border median, normalized by 255, and uses no CAI outcome. Complete candidate values make it retrospective and non-deployable. The registered protocol predates formal oracle execution; the comparator is not a damage-severity measure or a universal saliency definition. Appearance minus mechanical CAI AUEBC was 0.0070800594 (95% CI 0.0047993566–0.0097402930), with the CAI oracle favored in all six domains. Across 276 paired initial-state maps, mean Spearman agreement was 0.0222120 and mean top-decile overlap was 0.2003106.

Legacy reconstruction evidence is distinct from appearance saliency and is retained here without relabeling. The one-shot mechanical oracle had CAI AUEBC 0.013458 versus 0.017188 for the legacy reconstruction oracle; the mechanical improvement was 0.00373 (95% CI 0.00284–0.00469), favorable in all six domains (`U3_RECONSTRUCTION_ORACLE`). In the later registered cross-objective diagnostic, the CAI-task oracle improved CAI AUEBC by 0.04862 (95% CI 0.04527–0.05205) (`U4_ORACLE_CAI_SPECIFICITY`), while the reconstruction oracle improved normalized RGB MSE by 5.503×10^{-4} (95% CI 5.006×10^{-4} – 6.063×10^{-4}) (`U4_ORACLE_IMAGE_SPECIFICITY`). Source-trained global mechanics and reconstruction masks produced the predeclared support indicator 0, so the learned global mechanics mask did not reproduce that legacy oracle separation (`U4_LEARNED_SPECIFICITY_BOUNDARY`). These results remain bounded to the registered normalized-RGB-MSE objective.

Under identical strict-OOF states and splits, Ridge–Huber best-action agreement was 0.675 (95% CI 0.622–0.727) (`U5_RIDGE_HUBER_BEST_ACTION`), and top- k agreement was 0.685 (95% CI 0.650–0.719) (`U5_RIDGE_HUBER_TOPK`). Ridge–MLP best-action agreement was 0.233 (95% CI 0.182–0.283) (`U5_RIDGE_MLP_BEST_ACTION`), and top- k agreement was 0.213 (95% CI 0.192–0.235) (`U5_RIDGE_MLP_TOPK`). Because the shallow MLP was substantially less accurate, these contrasts do not isolate predictor architecture at equal predictive accuracy.

4 S4. State-conditioned valuation and source controls

The registered exact-budget set regrets were 0.08171 for the static scorer (`01_STATIC_SET_REGRET`), 0.07993 for the global score (`01_GLOBAL_SET_REGRET`), and 0.07978 for the random median (`01_RANDOM_SET_REGRET`). These source-only references annotate the static representation; they do not establish information-theoretic unobservability.

At the endpoint, the real-state trajectory recovered 26.6% (95% CI 10.6%–41.0%) of the static-to-independent-full-field error ratio (`03_FULL_FIELD_RECOVERY`). Its denominator is the independent internal-field sensitivity estimator, not the matched B-family field path.

The acquired-position/history control inherits the exact positions, ordering, and costs of the recipient trajectory while removing measured content. The shuffled-content control preserves recipient positions and budget while sourcing values from separated donor content.

An additional legacy reconstruction-derived control retains the frozen `reconstruction` mode and registered normalized-RGB-MSE objective on the same legal trajectory. Real minus this control was 0.03419 in CAI MAE (95% CI 0.02450–0.04384), with real content favorable in one of six domains (`03_REAL_MINUS_RECONSTRUCTION`). This technical control is not the appearance-saliency comparator and was not relabeled as saliency. These control semantics are required when interpreting the adverse real-minus-control directions.

Figure 1 provides a result-independent qualitative check across all domains. Within each sorted

domain, the lexicographically first specimen was selected without using an outcome, effect size, or direction.

5 S5. Cost-constrained realization and implementation diagnostics

The reachable-set analysis is restricted to the registered two-action pool at 6.25%. The joint near-oracle compares reachable sets retrospectively; it is not deployable and does not establish arbitrary-horizon planning regret.

For `A3_FEEDBACK_BENEFIT`, the no-feedback reference retained an AUEBC advantage of 1.496×10^{-5} (95% CI 1.064×10^{-5} – 1.944×10^{-5}). With the registered sign convention, feedback benefit was -1.496×10^{-5} (95% CI -1.944×10^{-5} to -1.064×10^{-5}), with feedback favorable in two of six held-out domains. This is a frozen implementation diagnostic and is not generalized to other state representations or policies.

For `A4_BASELINE_MINUS_MAVIS`, equal-domain CAI AUEBC was 0.125053 for the supervised state-conditioned implementation and 0.124992 for the static reference. Reference minus learned was -6.114×10^{-5} (95% CI -8.461×10^{-5} to -3.777×10^{-5}), and the learned implementation improved two of six held-out domains. The direction favors the reference. This comparison is one system-level endpoint and does not define the performance of the full Task- Relevant Information Acquisition framework.

6 S6. Provenance, chronology, and scope boundaries

The evidence chronology separates pre-existing spatial and static-reference analyses, the frozen outer implementation endpoint, and later diagnostic analyses. Later diagnostics reused fixed states, source bindings, splits, and outcomes; they were not used to select or modify the frozen endpoint. No model training, tuning, result selection, frozen scientific artifact, or existing canonical numerical value was changed. The controlled authority migration adds three claims from preregistered frozen appearance evidence and changes only paper visibility and narrative roles for legacy reconstruction claims.

The machine-readable visibility map assigns every canonical claim to main headline, main support, one main system diagnostic, or supplement-only use. Source-hash records bind every number to its immutable result artifact, and the supplement manifest independently records copied-file digests. External-data records are feasibility audits only and contain no external method-performance evaluation.

Scope remains the paired 276-specimen, six-domain CFRP data program; exact native-raster digital cost; the fixed predictor families, state representations, action hierarchy, and planners; and the registered CAI and appearance-saliency comparison. Legacy evidence retains the registered CAI and normalized-RGB-MSE tasks. Physical path length, coupling, settling, hardware timing, industrial cost, and external empirical transfer are not established.

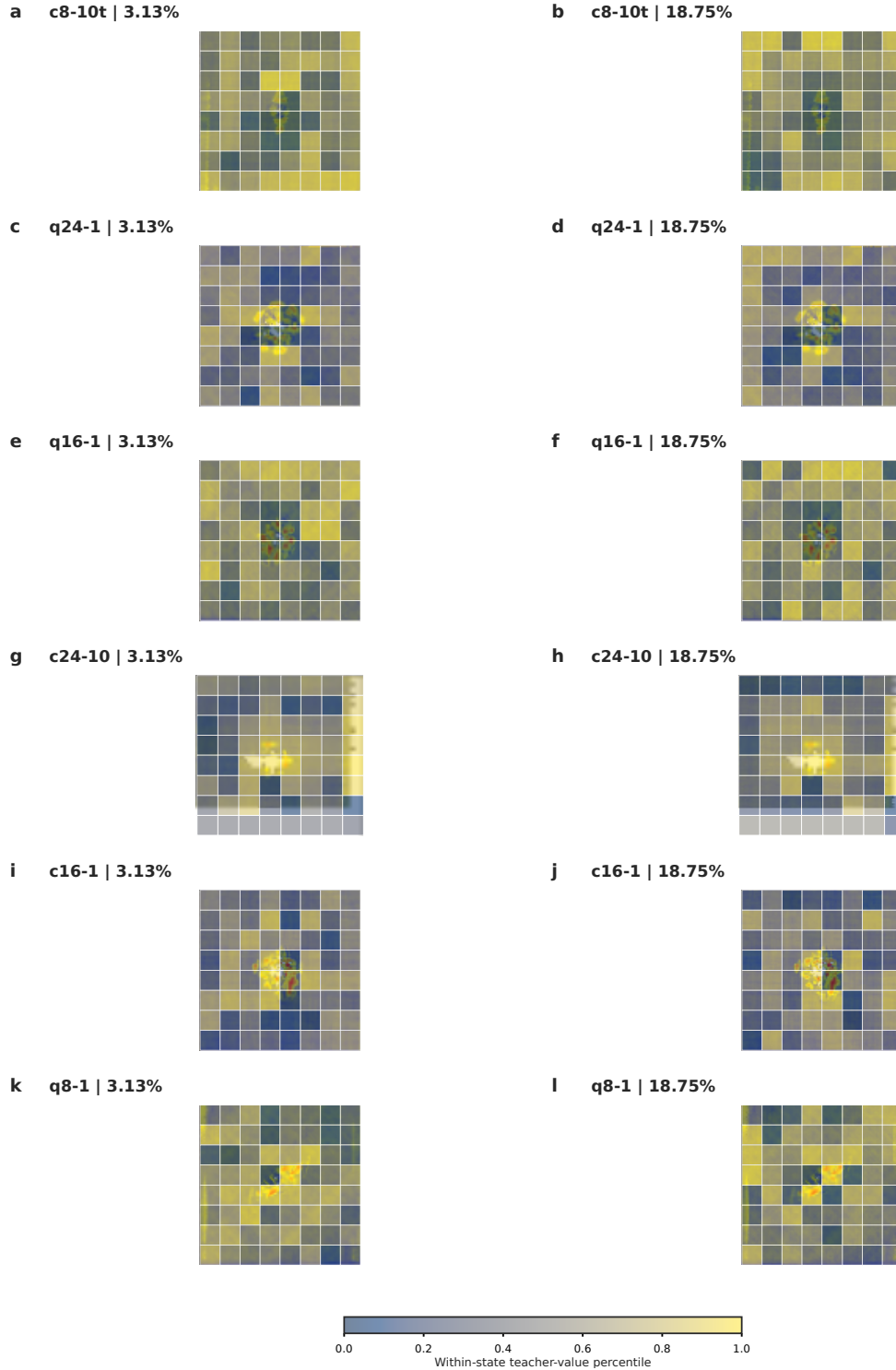


Figure 1: Cross-domain state-conditioned priority gallery. Each row shows one deterministically selected specimen from a different held-out domain. Left and right panels show the registered 3.13% initial and 18.75% states for the same trajectory. Overlays encode within-state strict-OOF teacher-value percentiles on the legal 8-by-8 action grid; they do not encode comparable absolute magnitudes across panels.